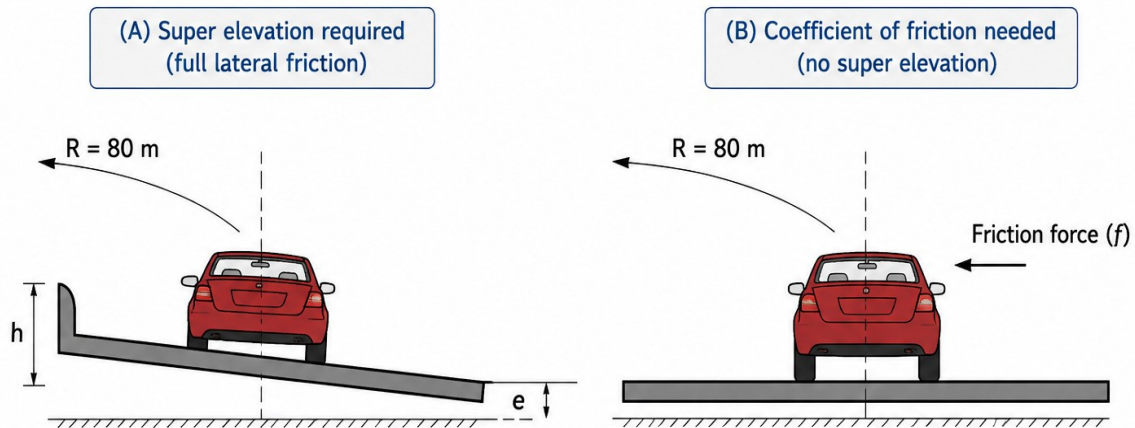


1. The radius of a horizontal curve is 80 m. The design speed is 45 kmph and design coefficient of lateral friction is 0.15. Determine the super elevation required if full lateral friction is assumed to develop and coefficient of friction needed if no super elevation is provided.

Answer:



Given,

$$R = 80 \text{ m}$$

$$V = 45 \text{ km/h}$$

$$f = 0.15$$

For horizontal curve,

$$e + f = \frac{V^2}{127R}$$

Super-elevation is required when full lateral friction develops.

$$f = 0.15$$

Therefore,

$$e + 0.15 = \frac{45^2}{127(80)}$$

$$e + 0.15 = 0.1992$$

Hence,

$$e = 0.1992 - 0.15$$

$$e = 0.0492$$

Therefore, the required super-elevation is:

$$e = 4.92\%$$

$$e = 1:20.3$$

Coefficient of friction when no super-elevation is provided i.e., $e = 0$

Thus,

$$f = \frac{V^2}{127 R}$$

$$f = \frac{45^2}{127(80)}$$

$$f = \frac{2025}{10160}$$

$$f = 0.1992$$

Therefore, $f \approx 0.20$

2. **How is implementation of bolsters help to stabilize the unstable slope? Describe with appropriate sketches the steps of implementation of bolster for purpose of stabilization of degraded slopes.**

Answer:

A bolster is a horizontal log of timber, bamboo, or brushwood structure placed across the face of a degraded slope and firmly anchored into the ground. A series of bolsters creates small terraces or steps on the slope.

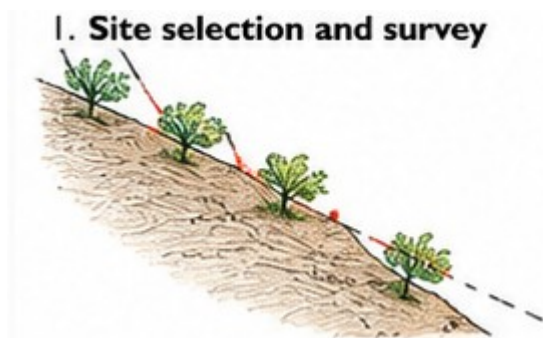
Bolsters stabilize slopes through the following mechanisms:

- i. **Reduction of slope length:** A series of bolsters divides a long slope into smaller sections, reducing the effective slope length.
- ii. **Reduction of runoff velocity:** The bolsters obstruct surface runoff and reduce its velocity.
- iii. **Control of soil erosion:** They prevent the downward movement of loose soil and surface debris.
- iv. **Promotion of vegetation:** The trapped soil provides a suitable medium for grasses, shrubs, and other vegetation to establish.
- v. **Root reinforcement:** Plant roots bind the soil particles and increase the shear resistance of the surface soil.
- vi. **Improved drainage and infiltration:** Properly installed structures can reduce concentrated surface flow and encourage infiltration.

The steps for implementation of bolsters are as follows:

i. **Site selection and survey**

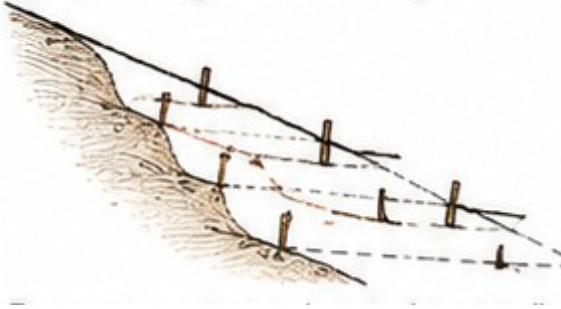
- Identify the degraded or unstable portion of the slope.
- Examine slope angle, soil condition, drainage, erosion, and landslide-prone areas.
- Mark the area requiring treatment.



ii. **Planning and marking**

- Mark the locations of the bolsters approximately along the contour of the slope.
- Determine suitable spacing according to slope steepness, soil condition, and site requirements.

2. Planning and marking



iii. Construction of trenches

- Dig shallow trenches along the marked contour lines.
- The trench should be sufficiently deep to provide firm seating for the bolster.

3. Cutting of trenches



iv. Placement of bolster

- Place logs, timber, bamboo, or bundled brushwood horizontally in the trenches.
- Maintain good contact between the bolster and the ground.

4. Placing the bolster



v. Anchoring

- Drive strong wooden or bamboo stakes into the ground at the ends and, where necessary, at intermediate points.
- Firm anchoring prevents the bolster from being displaced by runoff or soil movement.

5. Anchoring



vi. **Backfilling and compaction**

- Place excavated soil on the uphill side of the bolster.
- Compact the soil properly to provide firm support and encourage sediment accumulation.

6. Backfilling and compacting



vii. **Vegetation/planting**

- Plant suitable grasses, shrubs, and other vegetation above and behind the bolster.
- Vegetation helps bind the soil through its root system and provides long-term erosion protection.

7. Vegetation/planting



viii. **Maintenance**

- Install additional bolsters at suitable vertical intervals over the degraded slope.
- Regularly inspect the structures, repair damaged bolsters, clear excessive debris where necessary, and replace failed vegetation.

8. Repetition and maintenance



3. **State the basic differences between the suspension and suspended bridges. How are the loads transferred into foundation in a typical suspension bridge? Show all important elements of suspension bridge in a neat sketch.**

Answer:

The rugged and complex mountainous terrain of Nepal makes suspension bridges and suspended trail bridges an ideal solution for crossing rivers and deep valleys. The longest suspension bridge in Nepal is the Susta Suspension Bridge in West Nawalparasi, measuring 1,571 meters long and 1.6 meters wide.

The differences between suspension bridge and suspended trail bridge in design considerations and specific planning are listed below:

Suspension bridge	Suspended trail bridge
The bridge is designed to carry motor vehicles and heavy traffic.	This bridge is designed to carry pedestrian, livestock and cyclists.
They are of longer spans 100-200 m or more.	They are of short to medium spans 20-300 m.
Suspension bridge has wide deck of width 3.5-10 m or more for one or more traffic lanes.	Suspended trail bridge has narrow deck of 1-2 m suitable for pedestrian movement.
The design load considered in suspension bridges are heavy vehicular loads, impact loads, wind loads and seismic loads.	The design load considered in suspended trail bridge are pedestrian load, livestock load, wind and seismic loads.
The bridge requires strong and deep foundations.	The bridge requires comparatively lighter foundations and anchorages.
The deck has high stiffness that minimizes vibration and deflection.	The deck has moderate stiffness and limited deck movement.
The construction cost of suspension bridge is very high due to heavy structural components.	The construction cost of suspended trail bridge is comparatively low and economical.



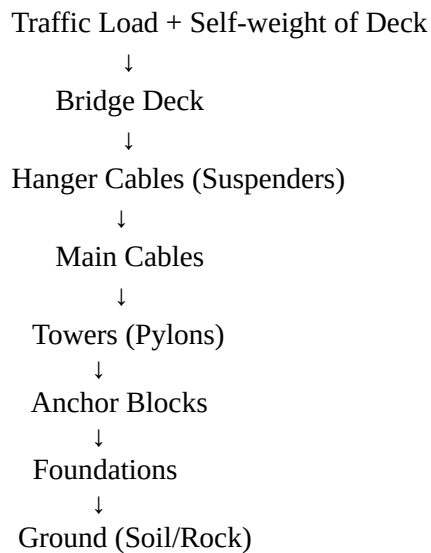
a. Suspension bridge



b. Suspended trail bridge

In a suspension bridge, the load from vehicles and the bridge itself is transferred to the ground through a sequence of structural members.

Load Transfer Mechanism in flowchart diagram:



- Bridge deck:
It carries traffic loads and transfers them to the hanger cables.
- Hanger cables:
Hanger cables also known as suspenders carry tension.
- Main cables:
Main cables also called suspension cables also carry tension.
- Towers:
They carry compression.
- Anchor blocks:
These blocks resist tension and prevent the main cables from pulling inward.
- Foundations:
Foundation of the bridge finally distributes all forces safely to the supporting soil or rock.

The main components of suspension bridge are given below:

i. Main cable (Suspension Cable):

- It is the primary load-carrying cable of the bridge.
- It passes over the tops of the towers and is anchored firmly at both ends.
- Usually made of thousands of high-strength steel wires bundled together.
- Carries the load of the bridge deck and traffic through the suspenders.
- Transfers the load to the towers and anchor blocks.

Function:

- Supports the entire bridge deck.
- Transfers loads to towers and anchorages through tension.

ii. Hanger cables (Suspenders):

- These are **vertical steel cables or rods** connecting the bridge deck to the main cable.
- Placed at regular intervals along the span.

Function:

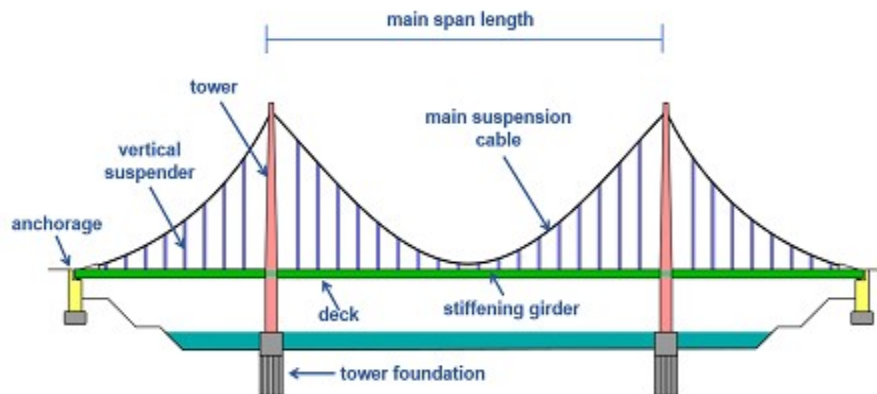
- Transfer the weight of the bridge deck and moving vehicles to the main cable.
- Keep the bridge deck suspended and properly aligned.

iii. Anchor cables:

- These are the portions of the main cable extending from the towers to the anchor blocks.
- They are firmly fixed into massive concrete anchorages.

Function:

- Balance the tension in the main cable.
- Transfer tensile forces safely to the ground through anchor blocks.
- Prevent the bridge from collapsing toward the center span.



4. In the treatment of $25 \times 10^3 \text{ m}^3/\text{day}$ of water, the amount of chlorine used is 15 kg/day . The residual chlorine after 10 min contact is 0.2 mg/lit . Determine the chlorine demand in mg/lit .

Answer:

Given,

$$Q = 25 \times 10^3 \text{ m}^3 / \text{day}$$

$$\text{Chlorine applied} = 15 \text{ kg} / \text{day}$$

$$\text{Residual chlorine} = 0.2 \text{ mg} / \text{L}$$

Chlorine demand is given by:

$$\text{Chlorine Demand} = \text{Chlorine Applied} - \text{Residual Chlorine}$$

$$\text{Chlorine applied} = \frac{15 \times 10^2 \text{ mg} / \text{day}}{25 \times 10^3 \text{ m}^3 / \text{day}} \times 25 \times 10^3 \text{ L} / \text{m}^3$$

Therefore,

$$\text{Chlorine Demand} = 0.6 - 0.2$$

$$\text{Chlorine Demand} = 0.4 \text{ mg} / \text{L}$$

5. Describe the various types of sewerage system. A sewerage system having the radius of 70 cm is laid with a gradient of 1 in 500. What will be the velocity of flow and discharge through the sewer when running one half full? Assume $N=0.012$ in Manning's formula.

Answer:

Separate and combined sewerage systems are the two common types of sewerage systems.

Separate system:

Separate system consists of two different sewers, one for convergence of sewage and another for rainwater.

It requires large areas since cleaning is not possible in small sewers.

If the rainfall is heavy and for short period of time and if sanitary sewage is required to be pumped, then separate system is provided.

Advantages:

- i. Load on treatment plant is reduced because waste type is taken separately.
- ii. Uniform process for purification.
- iii. Reduces the possibility of pollution due to overflow during excess rainfall.
- iv. Rainwater can be directly discharged into river.

Disadvantages:

- i. Requires two set of sewers, hence costly.

- ii. Occupies large area.
- iii. Choking problem can be seen in separate system.
- iv. There is a possibility of wrong connection due to confusion.

Combined system:

Combined system consists of only one sewer to carry both sanitary sewage and storm water.

Sewage must be treated before discharged into the streams.

It is suitable if street washings are as impure as sewage and must be treated before discharging.

Advantages:

- i. Combined system reduces the cost as it requires only one sewer.
- ii. It also reduces maintenance cost.
- iii. Strength of sewage is reduced by dilution.
- iv. House plumbing is easy and economic.
- v. Self-cleansing velocity of sewer can be easily achieved in combined sewerage system.

Disadvantages:

- i. Large dimensions of sewer are required in combined system.
- ii. Ventilation problem may occur.
- iii. Load on treatment plant increases in combined system.
- iv. Overflow of sewage during rainy season can occur in combined system.
- v. Handling and transportation of large size of sewer is difficult.

Given,

$$D = 70 \text{ cm} = 0.70 \text{ m}$$

$$r = \frac{D}{2} = 0.35 \text{ m}$$

$$S = 1500 = 0.002$$

$$n = 0.012$$

The sewer is running half full. Hence, for a circular sewer running half full:

$$A = \frac{\pi r^2}{2}$$

$$P = \pi r$$

Therefore, the hydraulic radius is:

$$R_h = \frac{A}{P} = \frac{\pi r^2 / 2}{\pi r} = \frac{r}{2}$$

$$R_h = \frac{0.35}{2} = 0.175 \text{ m}$$

Velocity of flow using Manning's formula:

$$V = \frac{1}{n} R_h^{2/3} S^{1/2}$$

Substituting the values:

$$V = \frac{1}{0.012} (0.175)^{2/3} (0.002)^{1/2}$$

$$V = 83.333 (0.175)^{2/3} (0.002)^{1/2}$$

$$V \approx 0.836 \text{ m/s}$$

Discharge through the sewer, for half-full circular sewer:

$$A = \frac{\pi (0.35)^2}{2}$$

$$A = 0.1924 \text{ m}^2$$

Discharge is given by:

$$Q = AV$$

$$Q = (0.1924)(0.836)$$

$$Q \approx 0.161 \text{ m}^3/\text{s}$$

6. **Describe the various renewable and non-renewable source of energy used for Nepal.**

Answer:

Nepal is one of the richest countries in term of natural resources. It has good potential for hydropower, solar and wind energy, while thermal power has a more limited role because of its dependence on imported fuel. The various renewable sources of energy used in Nepal are as follows:

i. **Hydropower:**

- a. It is the electricity generated by utilizing the energy of flowing rivers.
- b. Globally, hydropower covers about 14.3% of total electricity generation.
- c. Nepal has more than 6000 big and small rivers which has net electricity generation capacity of 83000 MW making it a second richest country in water resources after Brazil.
- d. Due to steep topography that provides the required head for power generation and availability of fast flowing rivers in Nepal, hydropower is most suitable option of alternate energy system.



ii. **Solar power:**

- a. It is the electricity generated by utilizing the sunlight.
- b. Powered by the photovoltaic effect (using silicon solar cells) or solar-thermal technology, it provides a renewable, clean, and increasingly cost-effective way to power homes, businesses, and entire utility grids.
- c. Solar energy covers about 10% of global electricity demand.
- d. It is ideal for remote mountain regions with high solar radiation where other alternate source of energy is difficult to supply or are expensive.



iii. Wind power:

- a. Wind energy is a renewable power source that utilizes the kinetic energy of moving air to generate electricity.
- b. Potential of wind energy exists in the district of Nepal with the presence of high velocity wind such as Mustang, Palpa, Dolpa, and other high-wind regions.
- c. It is currently limited in application.
- d. Can complement hydropower, particularly where wind availability differs seasonally from river flow.
- e. Suitable for small and decentralized power systems where site conditions are favorable.



iv. Thermal power:

- a. Can provide relatively reliable and controllable generation independent of rainfall.
- b. Useful as backup during hydropower shortages and emergencies.
- c. Can be located relatively close to major load centers.
- d. Construction can be faster than large hydropower projects.
- e. However, Nepal's dependence on imported fossil fuels makes it comparatively expensive and less environmentally desirable.



The various non-renewable sources of energy used in Nepal are as follow:

a. Petroleum products:

- Petroleum is one of the major non-renewable energy sources used in Nepal. It includes: Petrol, Diesel, Kerosene, Aviation fuel, LPG etc.
- These fuels are extensively used for transportation, cooking, industries, agriculture, and other purposes. Nepal depends heavily on imported petroleum products.

b. Coal:

- Coal is used mainly in:
 - Industries
 - Brick and cement production
 - Boilers and other industrial processes
- Its use produces significant amounts of greenhouse gases and air pollutants.

c. LPG:

- Liquefied Petroleum Gas (LPG) is widely used as a cooking fuel, particularly in urban areas.
- Although convenient and relatively cleaner than coal and traditional biomass during combustion, LPG is a fossil fuel and therefore non-renewable. Nepal imports most of its LPG.

d. Natural gas:

- Natural gas has comparatively limited use in Nepal because the country does not have significant domestic natural-gas production.
- It may be used in certain industrial applications, but Nepal relies largely on imported fossil fuels.

7. **Describe the factors, which should be considered in the construction of Hill irrigation canals. Also list the environmental aspects of hill irrigation.**

Answer:

Hill irrigation canals are constructed along sloping and often unstable terrain. Therefore, special consideration is required during their planning, design, and construction.

- a. Topography and Alignment
 - The canal should generally follow the contour of the hill to maintain a suitable and nearly uniform gradient.
 - Alignment should avoid very steep slopes, unstable areas, landslide-prone zones, and deep valleys as far as possible.
 - Excessive bends should be avoided.
 - The alignment should provide command over the maximum possible agricultural area.
- b. Geological and Soil Conditions
 - The stability and type of soil and rock along the alignment should be investigated.
 - Weak, fractured, or highly weathered rock should be avoided where possible.
 - Special measures are required in landslide-prone and erosion-prone areas.
 - The bearing capacity of the foundation should be adequate.
- c. Canal Gradient
 - The longitudinal slope should be sufficient to maintain the required velocity without causing excessive erosion.
 - Very steep gradients should be avoided because they may cause high flow velocity, scouring, and structural damage.
 - Drops and falls may be provided where necessary.
- d. Water Availability
 - The source should provide adequate and reliable water throughout the irrigation season.
 - Seasonal variation in river discharge should be considered.
 - The possibility of floods, sediment inflow, and drought should be assessed.
- e. Discharge and Canal Capacity
 - The canal should be designed for the required peak irrigation demand.
 - Allowance should be made for reasonable conveyance losses.
 - The canal capacity should be sufficient to serve the intended command area.
- f. Seepage and Water Losses
 - Hill canals may experience considerable seepage losses, particularly through fractured rock and permeable soils.
 - Canal lining may be provided where seepage losses are excessive.
 - Proper joints and construction quality should be maintained in lined canals.

The environmental aspects of hill irrigation are as follows:

The construction and operation of hill irrigation systems can produce both positive and negative environmental impacts.

Positive Environmental Aspects:

- i. **Increased agricultural productivity** through reliable irrigation.
- ii. **Improved food security** and livelihood of local communities.
- iii. **Multiple cropping** and better utilization of agricultural land.
- iv. **Vegetation development** due to improved availability of water.
- v. **Reduction in pressure on forests** where irrigation improves agricultural productivity and livelihood opportunities.
- vi. **Improved local socio-economic conditions** and employment opportunities.

Negative Environmental Aspects

- i. **Deforestation** due to canal construction and access roads.
- ii. **Soil erosion** caused by excavation, runoff, and leakage.
- iii. **Landslides and slope instability** due to hill cutting and water seepage.
- iv. **Waterlogging** in poorly drained areas.
- v. **Seepage-induced slope failure** where excessive water infiltrates unstable slopes.
- vi. **Sedimentation** of canals and downstream water bodies.
- vii. **Changes in natural drainage patterns** due to canal construction.
- viii. **Impacts on aquatic ecosystems** caused by diversion of water from natural streams.
- ix. **Loss of wildlife habitat and biodiversity** due to construction activities.

8. **Discuss about roles of construction materials and technology on strength of buildings. What are the local and modern construction materials and their implications in cost of buildings in Nepal?**

Answer:

Construction materials and construction technology play a major role in determining the strength, stability, durability, safety, and service life of a building. Proper selection of materials and appropriate construction techniques are particularly important in Nepal because of its diverse climate, difficult terrain, and high seismicity.

The roles of construction materials and technology on strength of buildings are given below:

- i. **Structural strength:**
Materials such as reinforced concrete, steel, masonry, timber, and engineered wood provide the required compressive, tensile, and shear strength to structural members.
- ii. **Durability:**
Good-quality materials resist deterioration caused by moisture, temperature changes, corrosion, fire, chemicals, and biological attack, thereby increasing the service life of buildings.
- iii. **Seismic resistance**
In earthquake-prone Nepal, the strength and ductility of materials are crucial. Properly designed reinforced concrete, structural steel, confined masonry, and properly detailed connections can improve earthquake resistance.
- iv. **Stability and stiffness:**
The stiffness and strength of materials influence the ability of a building to resist vertical loads, wind loads, seismic loads, and lateral deformation.
- v. **Fire and weather resistance**
Materials such as concrete, masonry, and suitable fire-resistant finishes can provide better protection against fire and adverse weather conditions.
- vi. **Quality control:**
The strength of a building also depends on the quality of raw materials and workmanship. Poor-quality cement, aggregates, bricks, steel, or improperly prepared concrete can significantly reduce structural strength.

Construction materials used in Nepal can broadly be classified into local/traditional materials and modern materials. The selection of materials directly affects the initial construction cost, transportation cost, labor cost, construction time, maintenance cost, and overall life-cycle cost of a building.

i. **Local Construction Materials**

Local materials are materials that are traditionally available or can be obtained relatively easily from the surrounding area. Major local construction materials in Nepal include:

- Stone
- Clay bricks

- Mud/earth
- Timber
- Bamboo
- Slate
- Thatch
- Lime
- Locally available sand and aggregates

These materials are commonly used in stone masonry, brick masonry, mud houses, timber structures, and traditional buildings.

Implications on Cost:

- **Lower transportation cost:**
Locally available materials require shorter transportation distances.
- **Lower material cost:**
Stone, earth, bamboo, and other locally available materials can be economical depending on location.
- **Lower equipment requirement:**
Traditional construction can often be carried out with simple tools.
- **Higher labor requirement:**
Traditional methods are generally labor-intensive and may increase labor costs and construction duration.
- **Variable quality:**
Natural materials may have considerable variation in strength and quality.
- **Maintenance cost:**
Some traditional materials may require more frequent maintenance.
- **Environmental benefit:**
Locally sourced materials can reduce transportation-related energy consumption and emissions.

ii. Modern construction materials:

Modern construction materials are manufactured or processed using improved technology and generally provide more uniform and predictable properties.

Major modern construction materials used in Nepal include:

- Cement
- Reinforcement steel
- Ready-mix concrete
- Concrete blocks
- Structural steel
- Aluminum
- Glass
- Engineered timber
- Waterproofing materials
- Thermal and acoustic insulation
- Fiber-reinforced materials
- Geosynthetic materials

These materials are widely used in modern reinforced-concrete framed buildings, steel structures, commercial buildings, and high-rise construction.

9. Discuss the present state of urban environment of Kathmandu valley. How do you think this situation can be made better?

Answer:

Kathmandu Valley is the major urban and economic center of Nepal. Rapid population growth, unplanned urbanization, increasing vehicles, inadequate infrastructure, and weak environmental management have created several environmental problems.

The three districts of Kathmandu Valley: Kathmandu, Lalitpur, and Bhaktapur has a combined population of approximately 3.03 million in the 2021 National Population and Housing Census. Kathmandu District alone had 2,041,587 people, Lalitpur had 551,667, and Bhaktapur had 432,132. The present urban environment can be discussed as follows.

i. Air Pollution

Air pollution is one of the major environmental problems of Kathmandu Valley. Major sources include:

- Vehicular emissions
- Road and construction dust
- Industrial emissions
- Open burning of waste
- Emissions from brick kilns and other sources

An ICIMOD study based on continuous monitoring at Khumaltar reported an annual average $PM_{2.5}$ concentration of approximately $49 \mu g/m^3$. This is nearly ten times the WHO annual guideline of $5 \mu g/m^3$.

The valley's topography can also restrict the dispersion of pollutants, particularly during periods of stagnant weather.

ii. Traffic Congestion and Noise Pollution

Rapid growth in the number of private vehicles has resulted in:

- Severe traffic congestion
- Increased fuel consumption
- Increased air pollution
- Noise pollution
- Loss of productive time

Poorly managed parking and inadequate road infrastructure further aggravate the problem.

iii. Solid Waste Management

Kathmandu Valley generates a large quantity of municipal solid waste every day. The problems generated in Kathmandu valley include:

- Inadequate waste segregation at source
- Irregular collection in some areas
- Open dumping
- Plastic waste
- Improper disposal of hazardous and electronic waste
- Pressure on landfill sites

These problems can cause unpleasant surroundings, odor, water pollution, and public-health risks.

iv. Water Pollution

The Bagmati, Bishnumati and other rivers of the valley have been heavily affected by untreated or inadequately treated wastewater, solid waste, and encroachment.

Major causes include:

- Direct discharge of sewage
- Industrial wastewater

- Solid waste dumping
- Surface runoff
- Inadequate wastewater treatment

More than 80% of Kathmandu's sewage has historically been reported to flow untreated into the Bagmati River, with an estimated discharge of about 150 MLD.

v. **Unplanned Urbanization**

Rapid and often unplanned expansion has resulted in:

- Conversion of agricultural land into built-up areas
- Encroachment on open spaces
- Development in environmentally sensitive areas
- Insufficient infrastructure
- Increasing building density
- Reduction of green areas

vi. **Poor Drainage and Urban Flooding**

Inadequate drainage capacity, encroachment of natural drainage channels, and extensive impervious surfaces can cause waterlogging and urban flooding during heavy rainfall.

The situations can be made better by:

- i. Improvement of public transportation and traffic management.
- ii. Control of air pollution by enforcing vehicle-emission standards.
- iii. Improvement of solid waste management by using 3R approach, waste segregation at the source etc.
- iv. Improvement of water management and promoting conservation of water resources.
- v. Promotion of planned urban development and sustainable development.
- vi. Development of green and open spaces in the urban areas.
- vii. Improvement of drainage and flood management systems.